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1. Smart Home automation using Intelligent agents

A smart home automation system receives inputs such as motion, temperature, time of day, light level and security sensor status. An intelligent agent uses these inputs to decide suitable actions.

(i) Simple Reflex agent:

A simple reflex agent follows if-then rules & does not remember previous situations

EX: If motion detected → turn ON light.

Advantages:

- Simple & fast
- Easy to implement.
- Requires less computational power.

Disadvantages:

Cannot learn user preferences
Cannot handle complex situations.
Makes decision only from current inputs.

2. Model Based Agent:

A model based agent maintains an internal state of the environment.

Ex: If the system knows that the user usually enters the bedroom at 10pm, it can prepare the bedroom lighting & temperature accordingly.

Advantages:

- Handles partially observable environments
- Uses past information

Disadvantages:

- More complex
- Requires memory & additional processing

3. Goal Based agent

A goal based agent chooses actions based on desired goals.

Possible goals are:

1. Maintain comfortable temperature
2. Reduce electricity consumption.

Ex: Goal: Maintain comfortable temperature

The agent observes that the temperature is 30°C and decides to turn on the AC until desired temperature is reached

Advantages:

- Can plan actions
- More flexible than rule based systems

Disadvantages.

- Finding the best sequence of actions can require more computation
- Goals alone do not indicate which of several acceptable solutions is best

4. Utility - Based Agent

A utility-based agent considers the quality of different possible actions and selects the one with the highest utility.

For ex: Comfort + Energy Saving + Security : Overall utility.

Suppose the temperature is high and nobody is in the room. Instead of immediately turning ON the AC the agent may keep it off to save energy.

Conclusion:

A hybrid agent is most suitable because a smart home is a dynamic environment. It must react immediately to emergencies while also learning preferences and making long-term decisions.

2. Robot finding an exit in an unknown maze.

A robot placed in an unknown maze does not initially know the layout. It must explore the environment and find a path to the exit.

Two useful uniformed search algorithms are:

- Uniform cost search (UCS)
- Iterative Deepening Search (IDS)

They are called uniformed because they do not use additional knowledge or a heuristic about where the exit

Uniform Cost Search (UCS):

UCS expands the node having the lowest path cost.

The robot can assign a cost to every movement.

For ex:

Start → A → B → Exit

Cost = 1+1+1 = 3

UCS continues exploring the cheapest paths until it reaches the goal.

Advantages:

- Complete: Finds a solution if one exists, assuming positive step costs.
- Optimal: Finds the least cost path.
- Useful when different moments have different costs.

Disadvantages:

- Can use a large amount of memory.
- Can be slow when many paths have similar costs.
- It explores based on cost, not closeness to the exit.

Iterative Deepening Search (IDS):

IDS repeatedly performs Depth-limited Search with increasing depth.

For ex: Depth 0 → Search
Depth 1 → Search
Depth 2 → Search
Depth 3 → Search.

The robot gradually increases it search depth until it finds the exit.

ex: Iteration 1: Search depth 0
Iteration 2: Search depth 1
Iteration 3: Serch depth 2
Iteration 4: Search depth 3

etc

Advantages:

- Requires much less memory than VCS
- Complete when the branching factor is finite
- Optimal when every action has the same cost.
- Useful when the solution depth is unknown.

Disadvantages:

- Repeats the search at smaller depths
- Can perform unnecessary work
- Not suitable for different action costs where optimal costs matter.

Best Combination:

For an unknown maze, the most effective choice would be: Model based + Goal Based Agent with IDS. The model based component allows the robot to remember explored locations & obstacles.